Environmental Studies

(CE 155)

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Learning outcomes

- Students have a deep commitment to protect the environment for sustainable development.
- Students must understand
  - The basic concepts of the environment
  - The laws of nature
  - Anthropogenic behaviours and impact on the environment
  - Mitigation measures

Reading Materials

- Visit Wikipedia for topics which are not too clear. Wiki has very good reference materials
- Living in the environment by George by Tyler Miller 2005
- Introduction to environment and society by J. P. Evans 2012

Environmental Studies

- An interdisciplinary field that includes both scientific and social aspects of human impact on the world.
- It involves an understanding of the scientific principles, impacts, economic influences and political actions.
- It looks at
  - Environmental problems
  - Human impact on the environment
  - Solutions to these environmental problems
  - Effects of natural and unnatural processes
  - Interaction of the physical components of the planet on the environment

Units

- The lectures have been divided into four units - Units 1, 2, 3 and 4
- Each unit contains the course content and there is assignment at the end for the student to work on
- Endeavour to do them since they give you the opportunity to assimilate and apply the course material to everyday life

Course Outline for the 4 Units

- Components of the environment, earth and the solar system
- Ethics and Sustainable development
- Environmental Resources
- Basic ecological concepts
- Matter and laws of energy
- Biogeochemical cycles
- Environmental crises
- Pollution and pollution control
- Contemporary environmental issues

Unit 1

Humans and nature – an overview

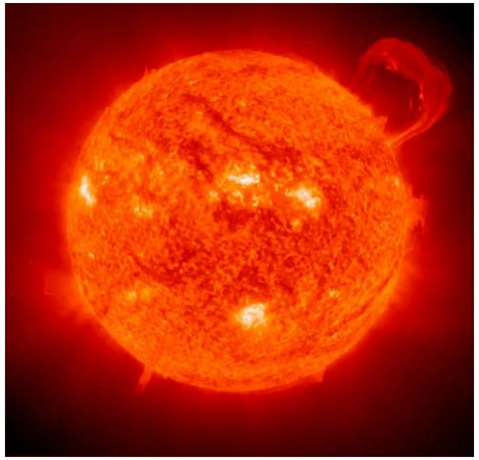
Topics

- Humans and the environment
- The solar system
- The structure of the Sun
- The electromagnetic spectrum
- Albedo and emissivity
- Our planet earth
- Other planets
- Theory of plate tectonics
- Earth's atmosphere
- Human societies and their impacts
- The Environment, Sustainability and Ethics

Introduction

- Environmental Science is broadly the protection of the environment through the study of sciences including natural and social sciences.
- Management of the resources for sustainability is key component of this course.
- To study our immediate environment the Earth and its cosmic environment must be understood
- Earth revolves around the Sun and it is the only planet where life is found.
- Failure of man to fit into the delicate structure is what is causing the environmental problems

The Sun and Solar System



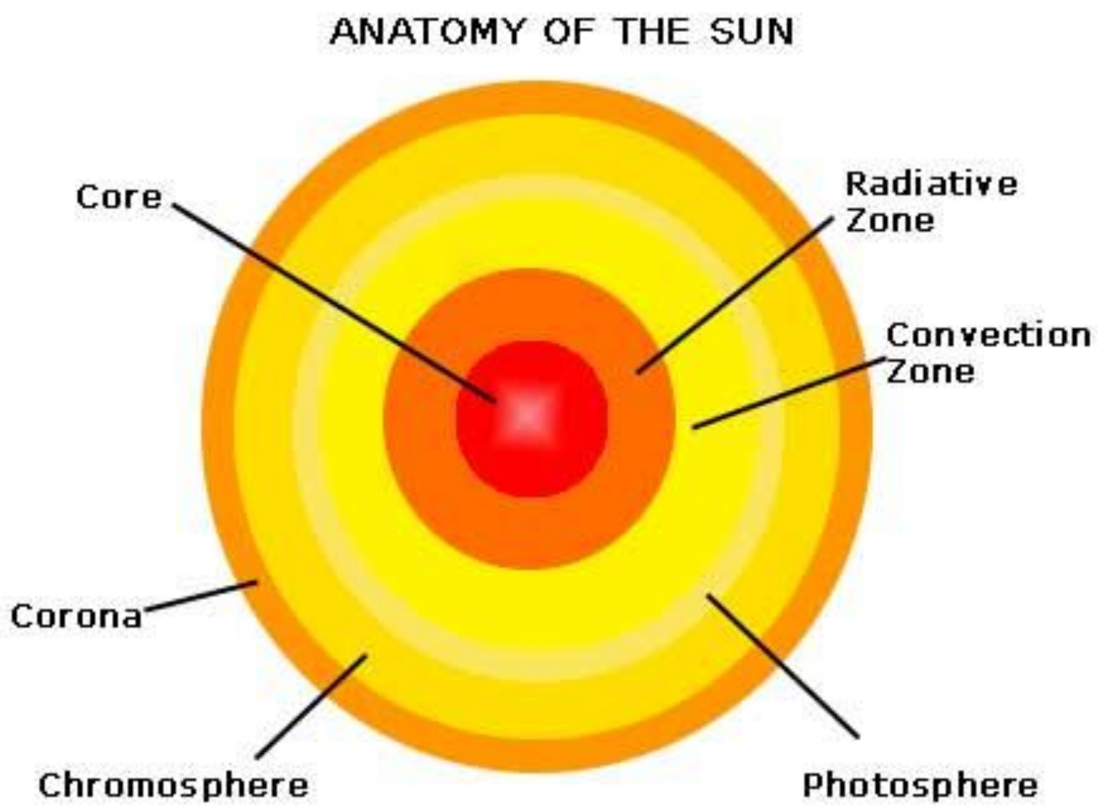
- Sun provides the earth with all the energy it needs
- Sun is one of the stars in the Milky Way Galaxy
- Hot gases make up the Sun; it is 1.35 million kilometres in diameter (about 109 times the size of the earth but 4 times lighter in weight than the earth) and 150 million kilometres from the Earth
- Observation and facts about the sun were studied through X-ray telescopes
- The Sun is a solar fusion reactor, which constantly converts Hydrogen atoms to Helium atoms
- The process involves a loss in mass (m) which is converted into energy in accordance with Einstein's law:

Energy from the sun = mc^2

c is the speed of light (3×10^8 m/s)

Structure of the Sun

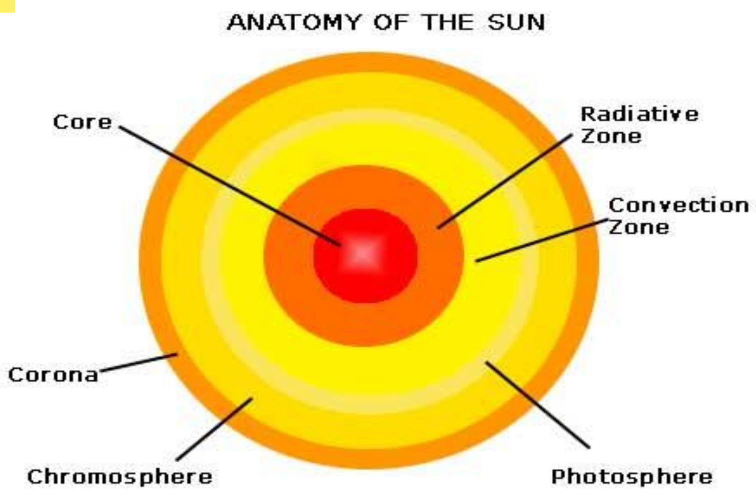
- The sun has different layers, namely-
- The inner layers are the:
  - Core, radiative and convective zones
- The outer layers which constitute the sun's atmosphere are the:
  - Photosphere, chromosphere and corona
- These layers have different temperatures



Structure of the Sun

Core

- Innermost part of the sun where the energy is produced
- Temperature more than a 15,000,000 ˚C
- Cannot be seen



Radiative zone and Convective zone

- Energy from the core is transferred to other regions
- via radiation in the radiative zone and convection in the convective zone.

Photosphere

- Is visible and it gives off light
- Temperature is about 3,700 to 6,000˚C

Structure of the Sun

Chromosphere

- The red colour is caused by the glow of hydrogen
- The temperature is 3700 - 7700˚C

Corona

- Outer part of the sun's atmosphere
- The average temperature is 1,000, 000˚C

- Chromosphere and Corona are visible only during a total solar eclipse
- The sun has cooler spots known as sun spots which appears dark on the photosphere.

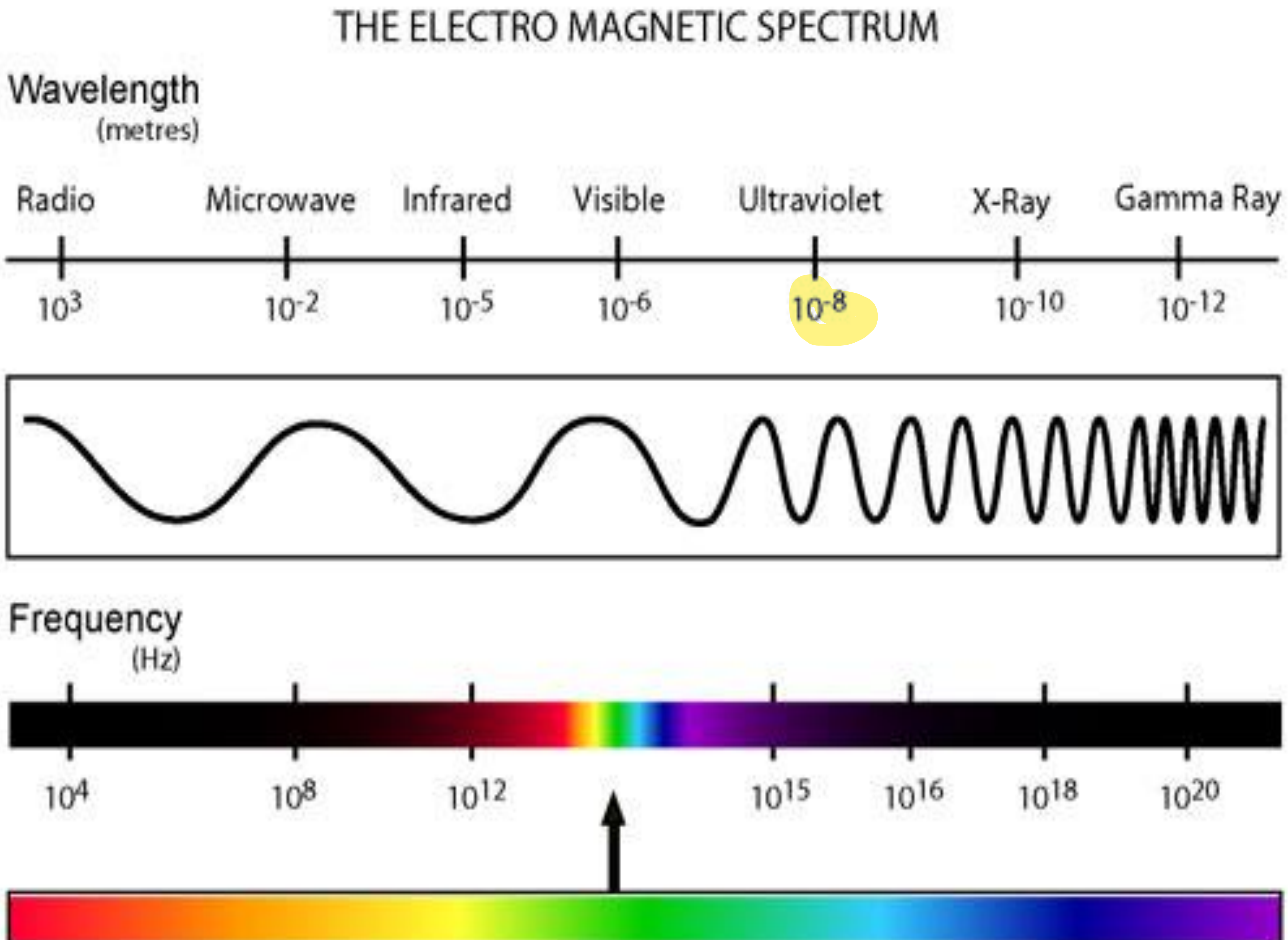
Importance of the sun

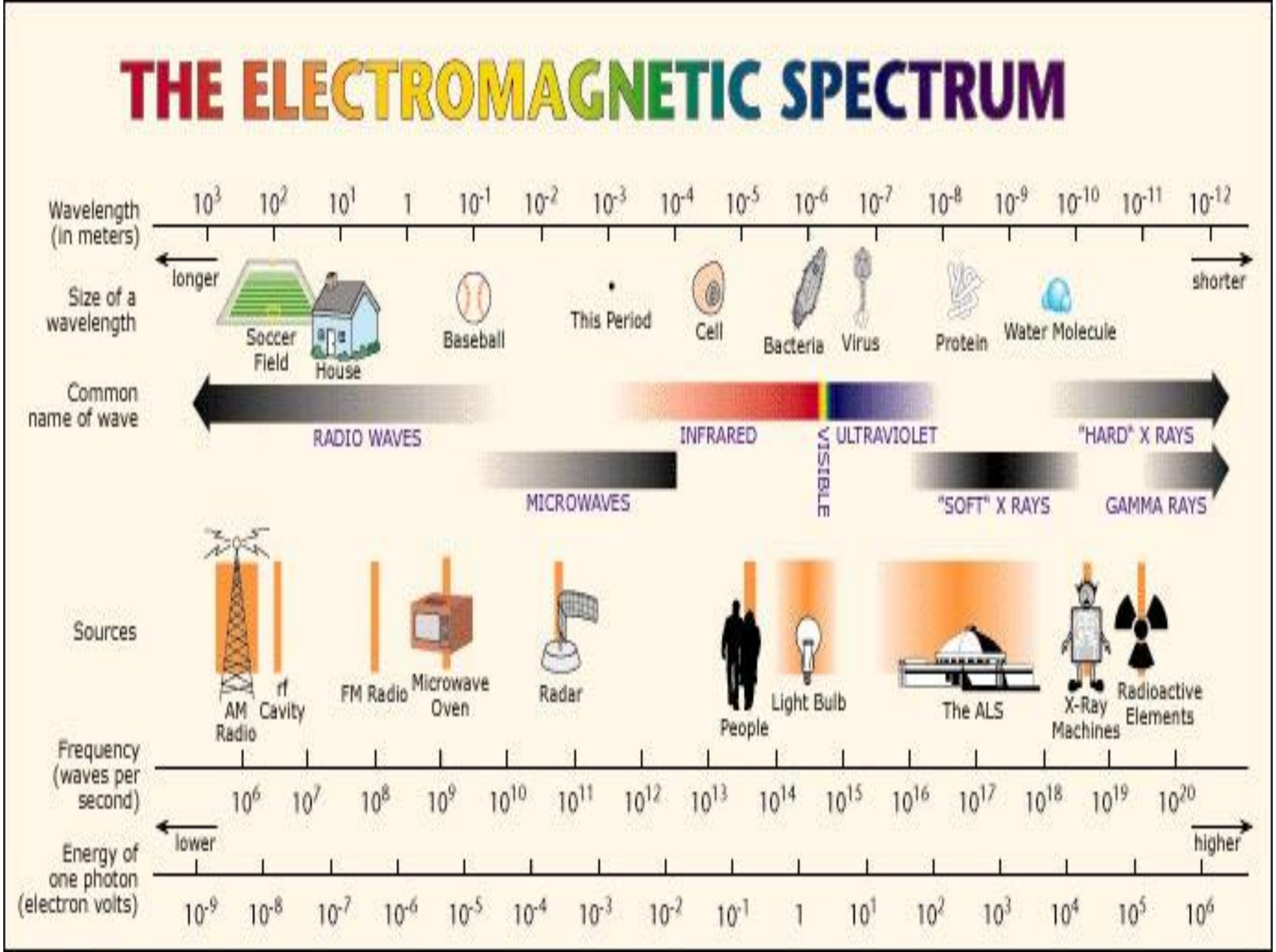
- The sun
  - keeps the earth warm. Without the sun, the earth's atmospheric temperature would be about 38°C less than what it is now
  - gives the earth light
  - controls the weather
  - provides energy for photosynthesis
  - provides vitamin D.

Solar electromagnetic spectrum

- Energy from the sun that is solar radiation reaches the earth in the form of electromagnetic waves.
- These waves according to their different wavelengths and frequencies constitute the electromagnetic spectrum. The different waves are:

i. Visible light rays
ii. Gamma rays
iii. X-rays
iv. Ultraviolet (UV rays)
v. Infrared
vi. Microwaves
vii. Radio waves

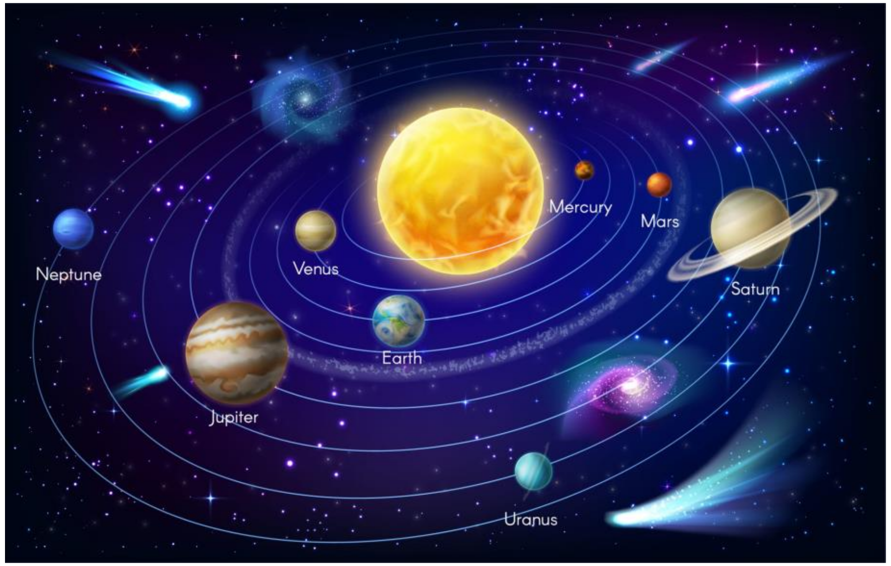

THE ELECTRO MAGNETIC SPECTRUM
Wavelength
(metres)
Radio
Microwave
Infrared
Visible
Ultraviolet
X-Ray
Gamma Ray
10^3
10^{-2}
10^{-5}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
Frequency
(Hz)
10^4
10^8
10^{12}
10^{15}
10^{16}
10^{18}
10^{20}



THE ELECTROMAGNETIC SPECTRUM
Wavelength (in meters)
10^3 10^2 10^1 1 10^-1 10^-2 10^-3 10^-4 10^-5 10^-6 10^-7 10^-8 10^-9 10^-10 10^-11 10^-12
longer
shorter
Size of a wavelength
Soccer Field
House
Baseball
This Period
Cell
Bacteria
Virus
Protein
Water Molecule
Common name of wave
RADIO WAVES
INFRARED
VISIBLE
ULTRAVIOLET
"HARD" X RAYS
MICROWAVES
"SOFT" X RAYS
GAMMA RAYS
Sources
AM Radio
rf Cavity
FM Radio
Microwave Oven
Radar
People
Light Bulb
The ALS
X-Ray Machines
Radioactive Elements
Frequency (waves per second)
10^6 10^7 10^8 10^9 10^10 10^11 10^12 10^13 10^14 10^15 10^16 10^17 10^18 10^19 10^20
lower
higher
Energy of one photon (electron volts)
10^-9 10^-8 10^-7 10^-6 10^-5 10^-4 10^-3 10^-2 10^-1 1 10^1 10^2 10^3 10^4 10^5 10^6

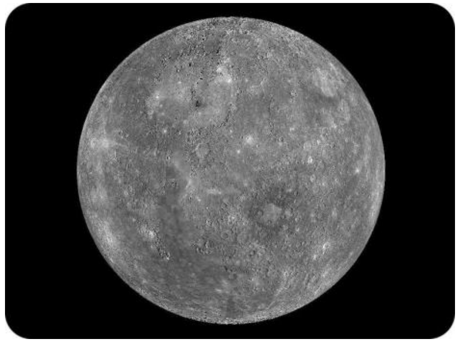
The Planets

- A planet is a celestial body orbiting a star that is massive enough to be rounded by its own gravity, is not massive enough to cause thermonuclear fusion, and has cleared its neighbouring region of planetsimals (IAU, 2006).
- Become gravitationally dominant, and there are no other bodies of comparable size other than its own satellites or those otherwise under its gravitational influence
- There are 8 planets.



Temperatures on the Planets

Planet Mercury



- It is about 58million km from the sun and the nearest to the sun
- Has no atmosphere due to low gravity and high day time temperature.
- Temperature during the day is 400°C and at night -200°C

Planet Venus

- It is 108 million km from the sun; has thin pale yellow clouds
- Atmosphere mainly contains $C0_2$ with 1 to 3% Nitrogen
- Has small amounts of helium, argon and water vapour and allows only 25% of the sun's rays to penetrate its surface and heat the rock.
- It is the hottest planet. Temperatures are around 475°C.

Temperatures on the Planets

Planet Earth

- It is about 150 million km from the sun
- Covered by 71% water, the remaining being land.
- Highest temperature is 57°C and coldest -89°C

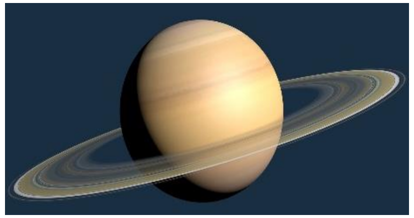
Planet Mars

- About 249 million km from the sun
- Atmospheric daytime temperature ranges from 0 to 27°C because of its CO_2
- Night time temperature is about –50 to150°C because of its thin atmosphere
- Has two moons.
- Also known as Red planet

Temperatures on the Planets

Planet Jupiter

- More than 778 million km from the sun, has clouds and 80 moons
- Has hydrogen, helium and methane, water droplets and ammonia crystals in it's atmosphere; radiate more heat back into space than it get from the sun
- Cloud top temperature is about -150°C.



Planet Saturn

- About 1.434 billion km from the sun. Also known as the ringed planet.
- About 83 known moons
- 96.3% of hydrogen , 3.25% of helium, then methane, ammonia and ethane
- Average temperature is about –138°C

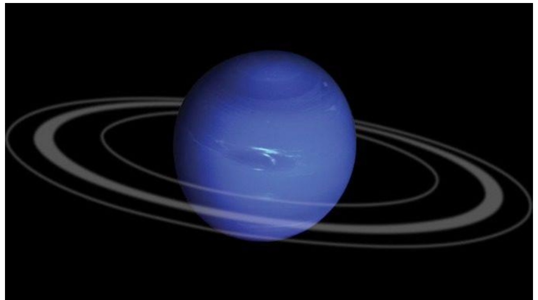
Temperatures on the Planets

Planet Uranus

- More than 2.9 billion km from the sun and has 27 moons
- Has 83% of hydrogen, 15% of helium and methane, hydrogen sulphide. Also water and ammonia ice.
- Cloud top temperature is about -224°C.

Planet Neptune

- About 4.5 billion km from the sun.
- 14 known moons. 80% of hydrogen, 19% of helium, then methane, ethane, ammonia and water
- Average temperature is about –200°C



The Earth's Moon

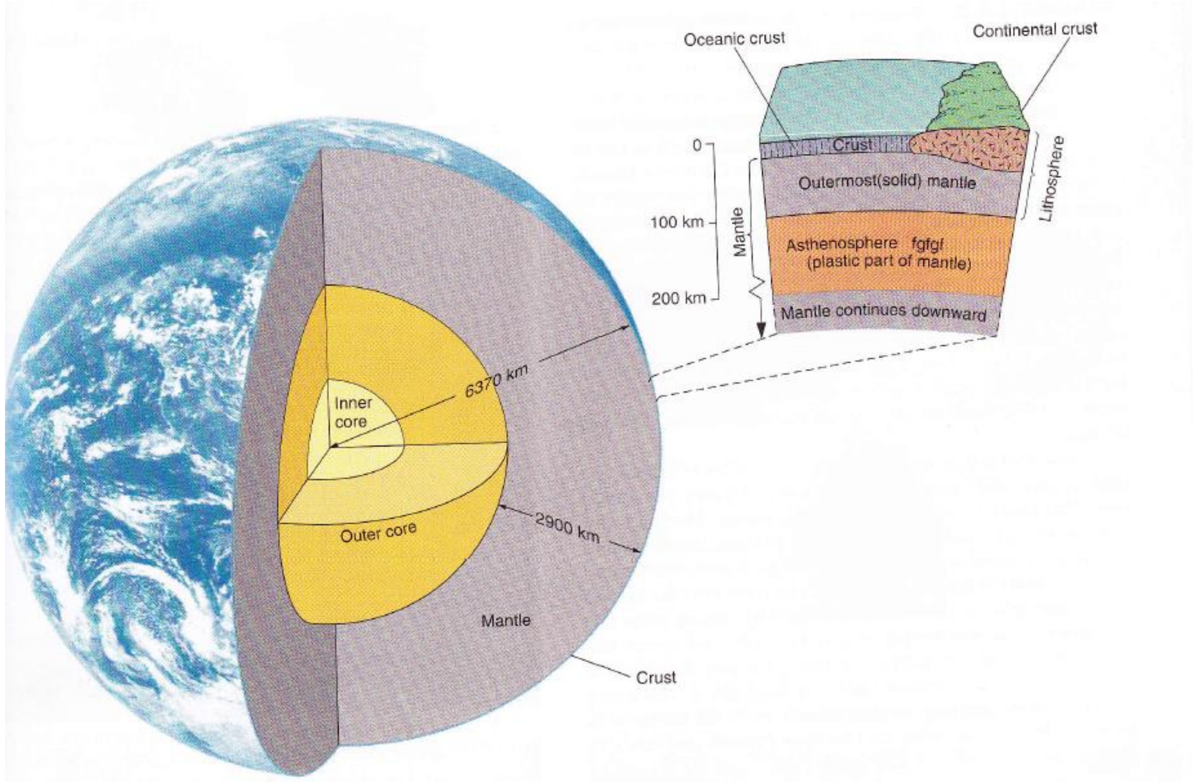
- The moon is a spherical rocky body which revolves round the earth; the earth's only natural satellite.
- It is about 385000km away from the earth.
- Has no atmosphere and therefore no weathering takes place
- No wind and erosion
- The moon takes about 27.3 earth days to orbit the earth.
- Temperature is 107°C at daytime and -153°C at night time
- The moon causes tides and thus influences activities of aquatic animals (e.g. mating and spawning time),
- Gives light to earth at night by reflecting the sun's light.
- Can be used to tell the time.

Our Planet Earth

- The planet earth is often called the living planet
- The earth is 12,800 km in diameter;
- Earth's atmosphere has large amounts of oxygen and water vapour; help to make life possible on earth
- Constantly revolves around the sun in an orbit once in 365 days in a year
- About 75% of the earth's surface is covered with water, 97.5% of the water is Saline
- Remaining 2.5% is found in surface water bodies and the rest tied up in icecaps, glaciers, in the earth and the atmosphere
- Only 0.32% of the world's water (surface water bodies) is available for use

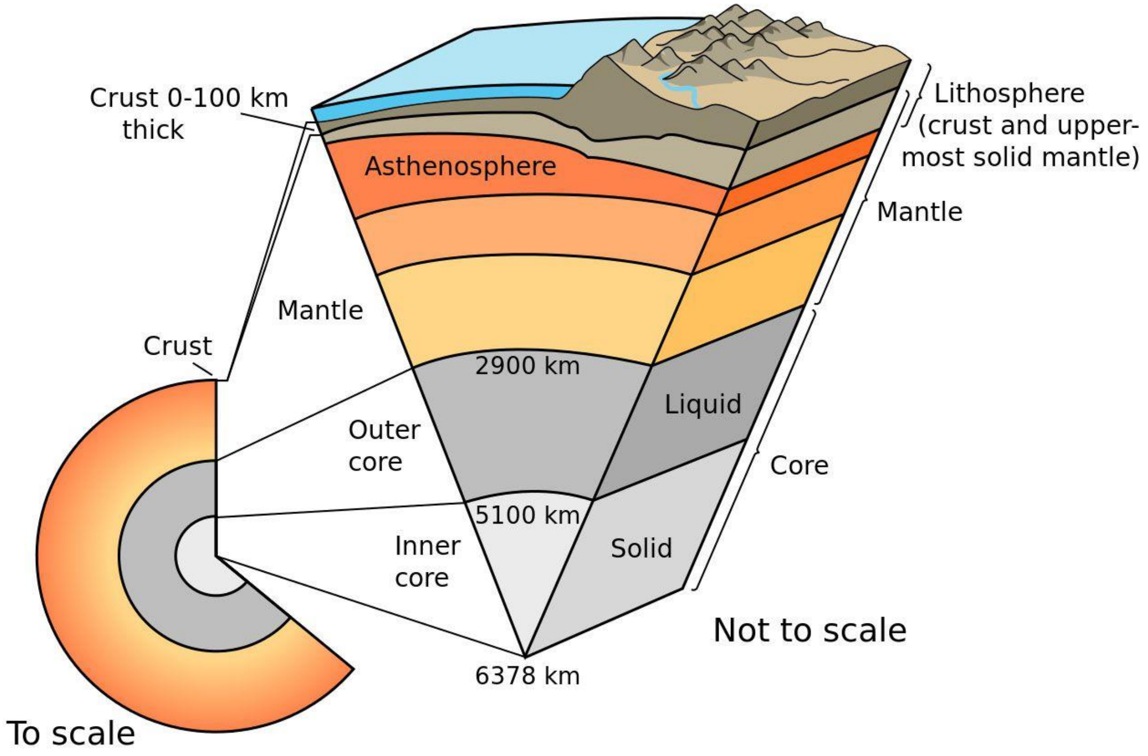
Structure of the earth

- Based on chemical composition, the earth is made up of the core, mantle and crust.
- The crust is the outermost, thin and less dense layer of the earth.
- Made up the continental crust (granite) and oceanic crust (balsalt).
- About 5km deep in oceanic crust, 30-40km deep in continental crust and about 100km deep in mountainous areas.



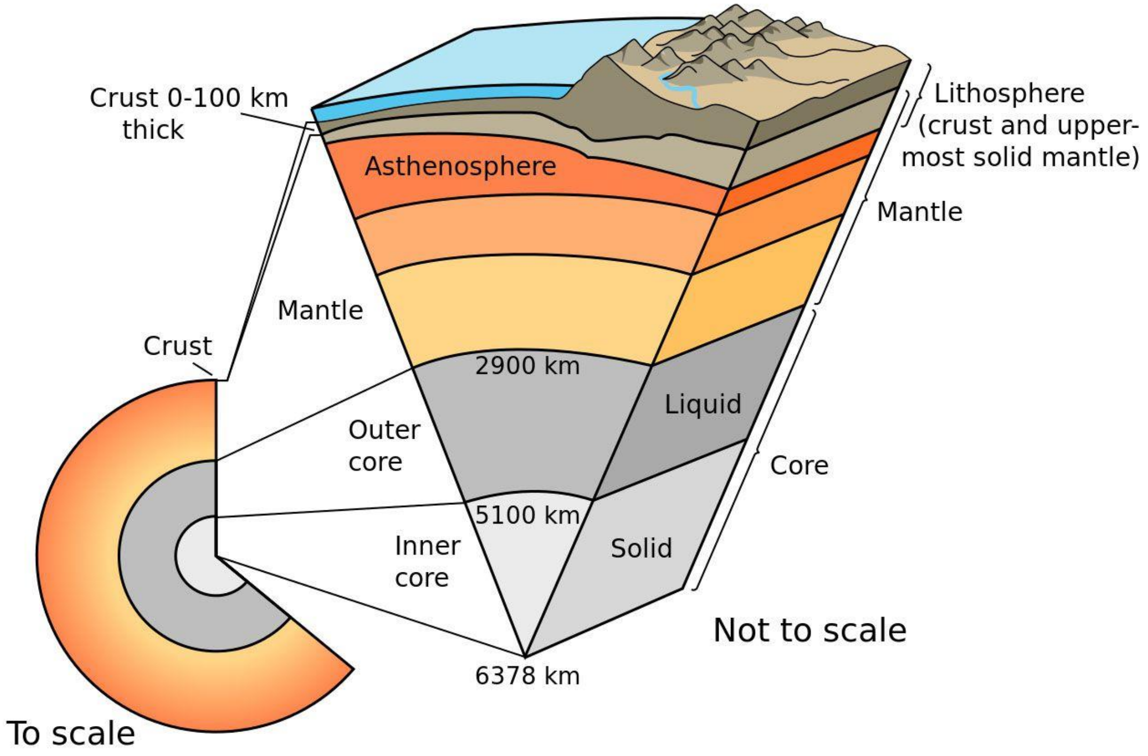
Structure of the earth

- The mantle is the layer extending from the base of the crust to the upper layer of the core.
- It is made up of dense, hot semi-solid rocks due to the occurrence of high temperatures and pressures
- About 2900km in thickness.
- Rock type is peridotite - rich in iron, magnesium and calcium.



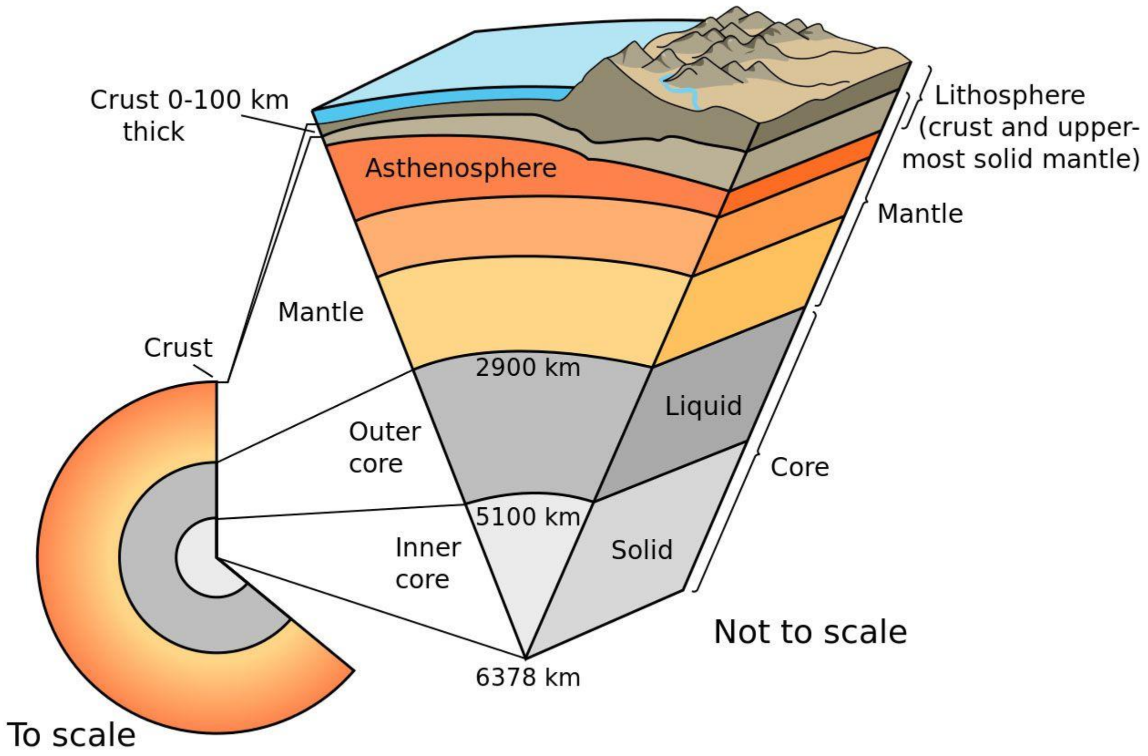
Structure of the earth

- The core lies in the centre of the earth.
- It is denser than the mantle because it is metallic (iron and nickel)
- Has two divisions
- The outer core which is liquid (2200km thick)
- The inner core which is solid (1250km)



Structure of the earth

- Based on mechanical properties, the earth can be divided again into
- Lithosphere – extends from the crust to the upper mantle.
- Made up of brittle, rigid rock materials. Thus, stresses can cause these rocks to break as seen in during earthquakes
- Asthenosphere – occurs below the lithosphere into the inner mantle
- Made up of hot, semi-solid material, viscous and capable of flowing.



Continental Drift Theory

- An earlier theory by Alfred Wegener in 1912 postulated that:
- all the continents were once together and drifted apart on the planet over time
- He observed the coast of West African and eastern South America can fit together
- Later, the theory of Plate tectonics was developed which showed that tectonic plates were moving not just continents

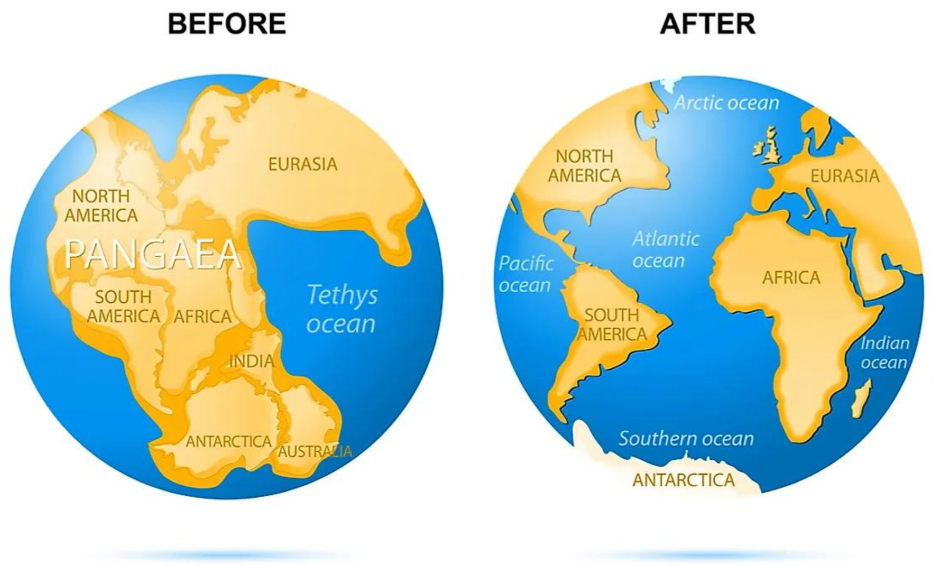
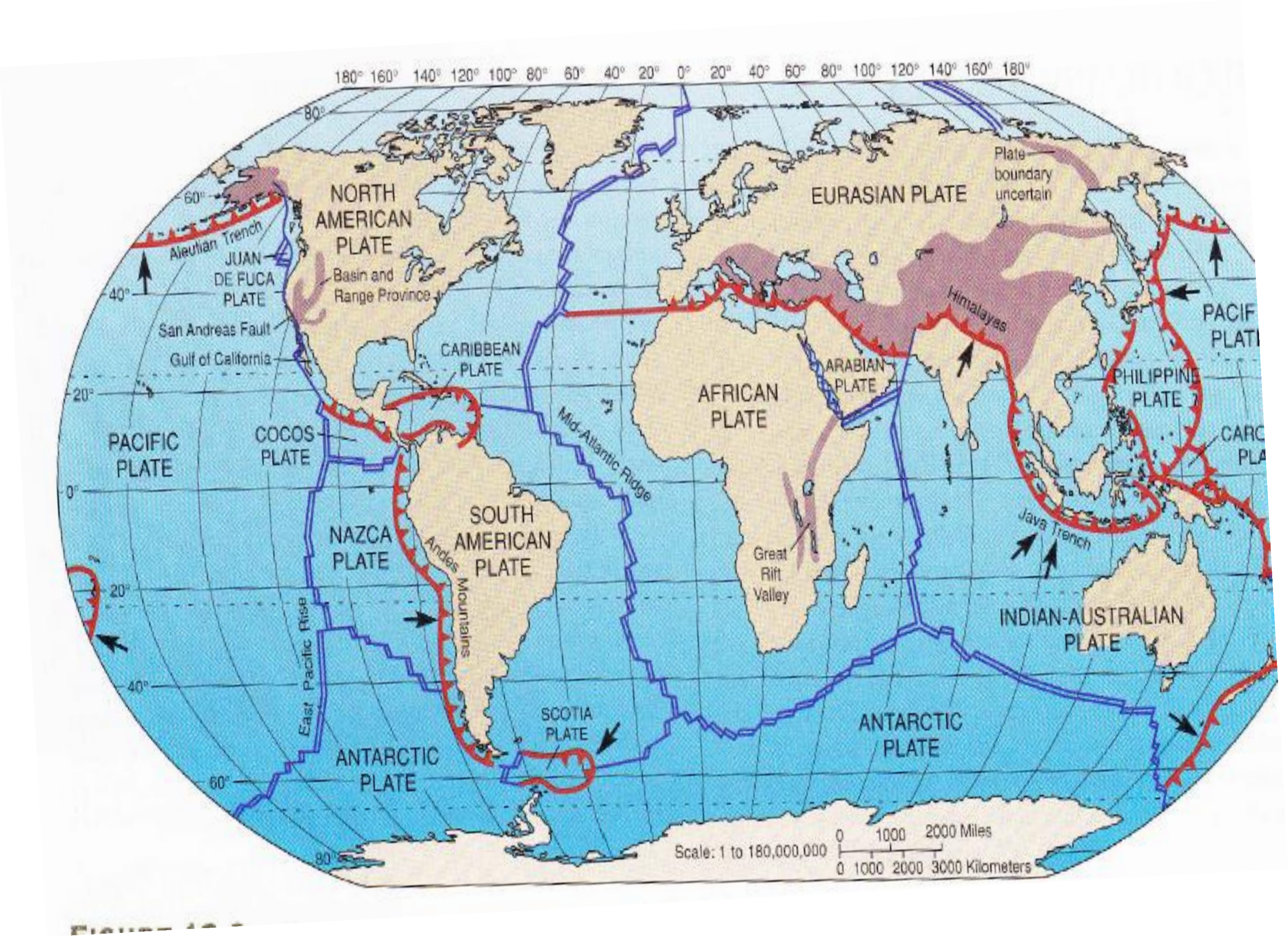


Plate Tectonics

- Plate tectonics is the scientific theory that the lithosphere is made up of large tectonic plates which glide over the asthenosphere.
- These movements result in the formation of mountains, floods, tsunamis, volcanoes and earthquakes.
- The lithosphere is divided into 7 large slabs of solid rock and several other small ones termed as tectonic plates.
- The large plates include the:

- Pacific Plate
- South American Plate
- African Plate
- Australian Plate
- North American Plate
- Eurasian Plate
- Antarctic Plate

Location of the Lithospheric plates



Natural Hazards 1 : Plate Tectonics

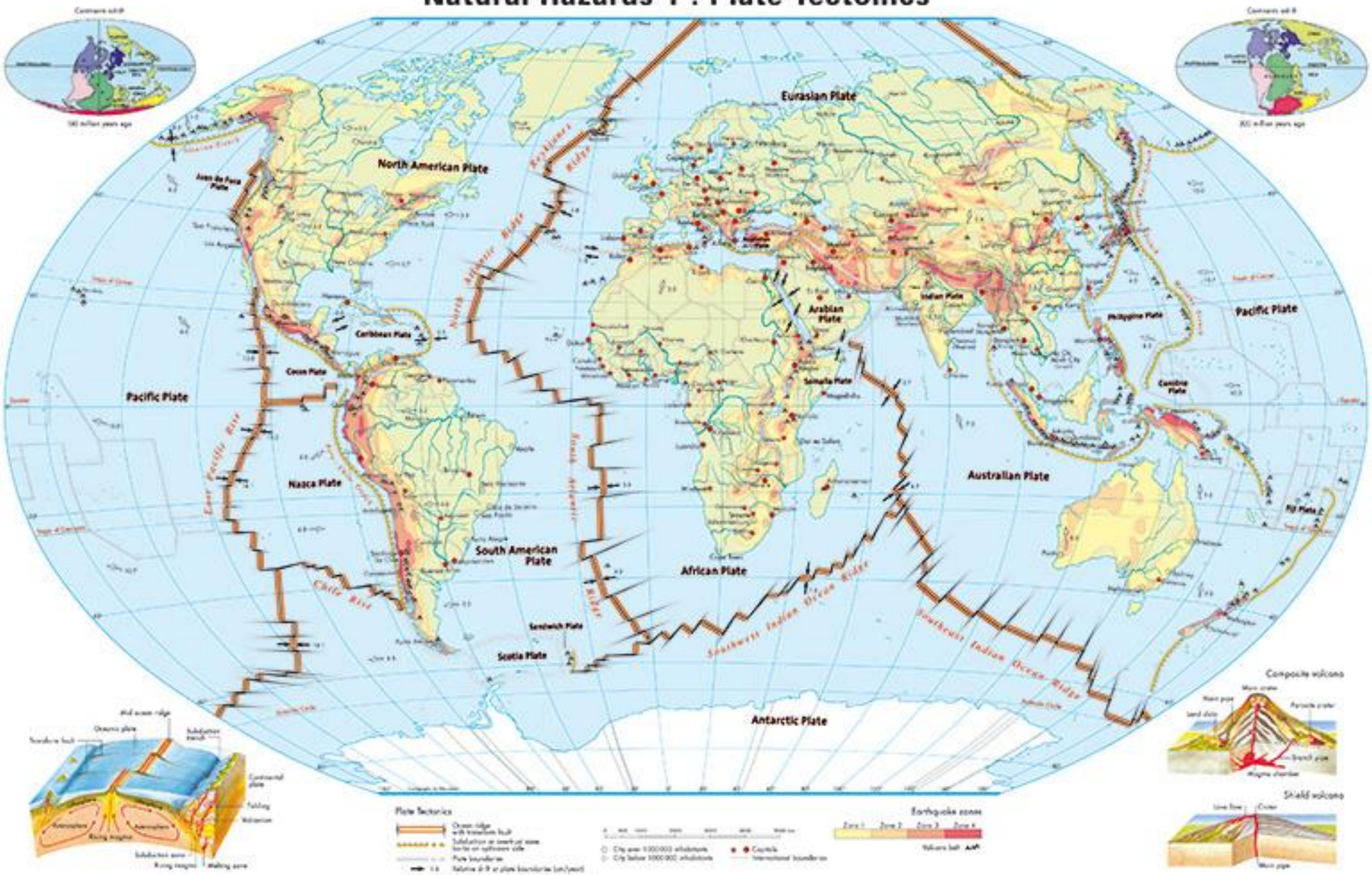


Plate Tectonics

There are 3 types of tectonic plate movements:

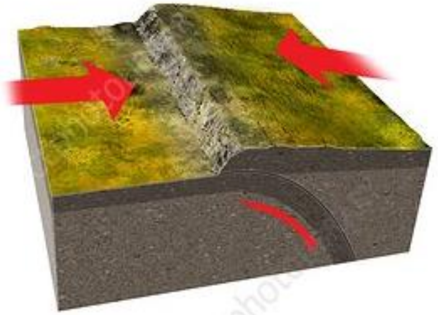
a. Convergent – when plates move towards each other to either form a

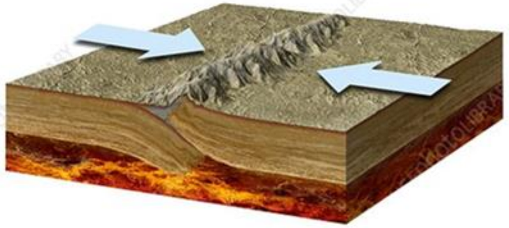
i. subduction zone (one plate slide under the other) to create earthquakes and volcanoes or

ii. continental collision to form mountain ranges

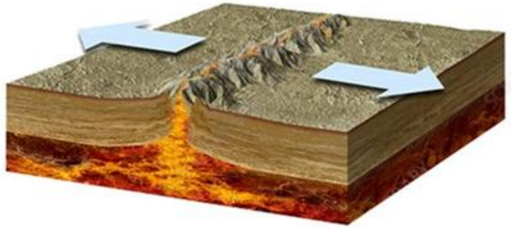
a. Divergent – when plates pull away from each other. This may lead to formation of rift valleys, volcanic islands, sea floor spreading

b. Transform- when plates slide or grind pass each other. This can lead to earthquakes, crustal deformation and lateral displacement of rocks





Convergent

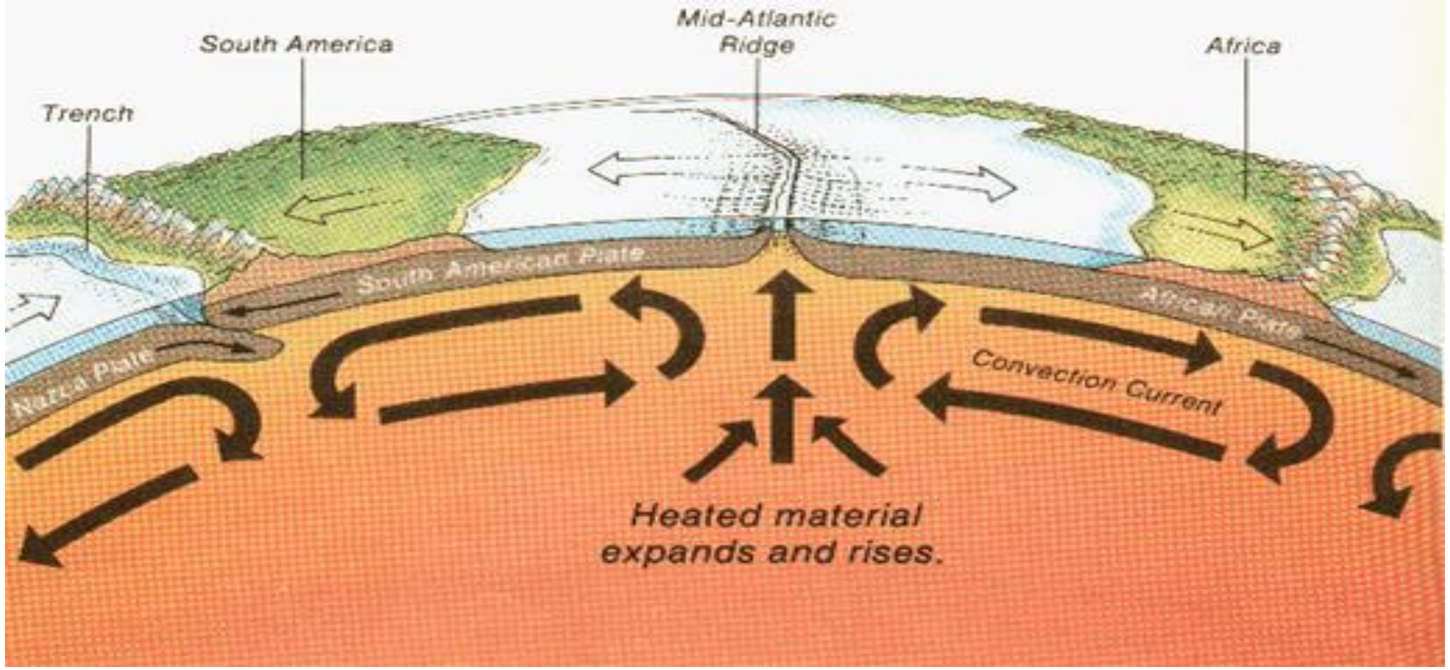


Divergent

Transform

Driving force of Plate Tectonics

The driving force behind plate tectonics is convection in the mantle. Hot material near the Earth's core rises, and colder mantle rock sinks.



https://www.britannica.com/science/plate-tectonics/Island-arcs
https://opengeology.org/textbook/2-plate-tectonics/

Earth's atmosphere

The atmosphere of the earth is the thin air surrounding the earth and it is composed of different layers depending on temperature.

The air which is a mixture of gases is retained in the earth by the force of gravity.

The atmosphere makes life possible on earth by

- Providing air for living organisms
- Shields organisms from harmful ultraviolet radiations emanating from the sun by absorbing them.
- Warms the earth's surface through greenhouse effect.
- Prevents extreme temperature differences between daytime and night time.
- Creates the pressure which enables water in the liquid state to exist on earth.

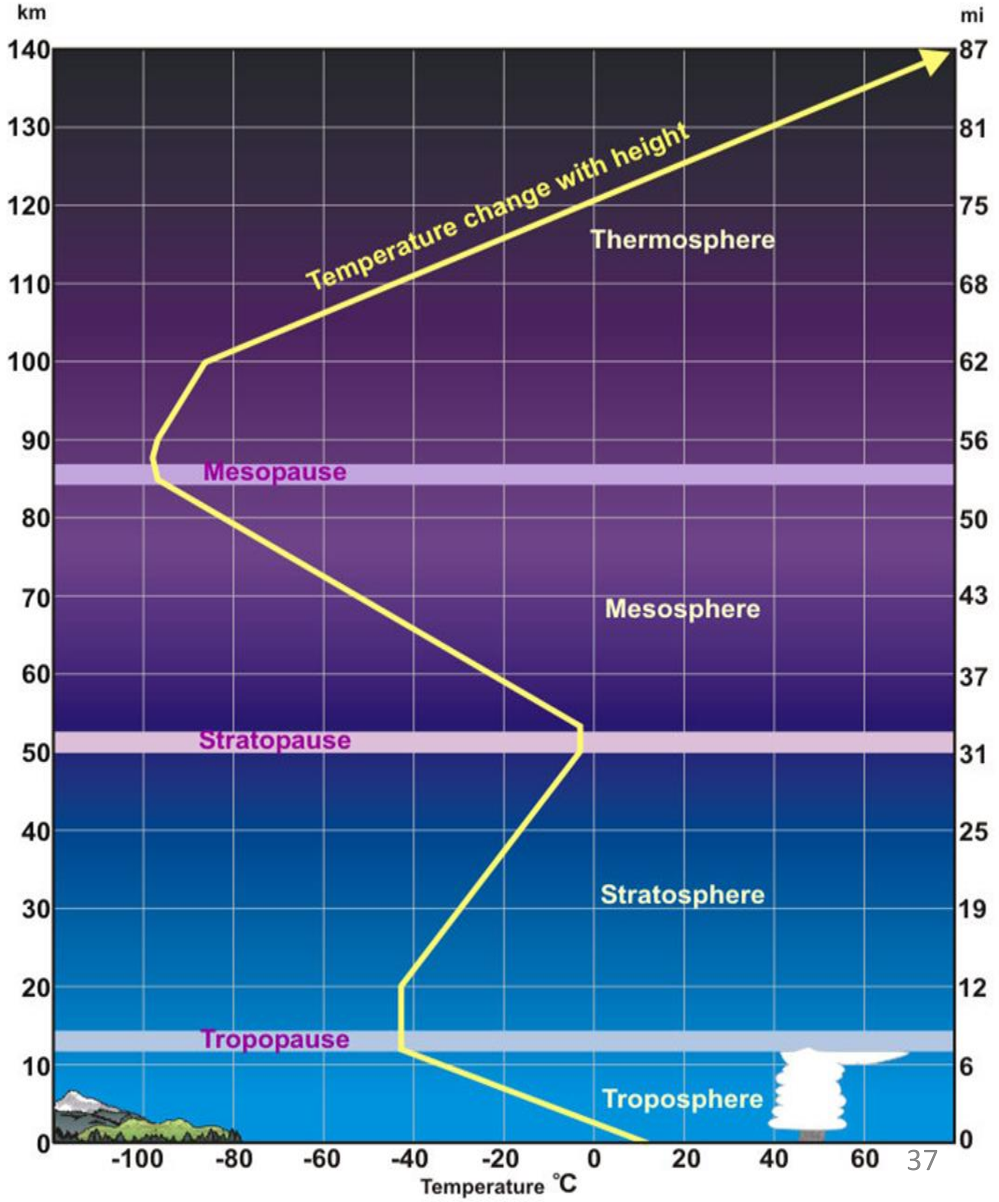
COMPOSITION OF AIR

The average percentage (%) and chemical composition of the air is as follows:

Nitrogen	78	Hydrogen (H_2)	5.0×10^{-5}
Oxygen	21	Xenon(Xe)	8.0×10^{-6}
Argon	0.9	Ozone (O_3)	2.0×10^{-6}
Carbon dioxide	0.03	Ammonia (NH_3)	6.0×10^{-7}
Neon (Ne)	0.0018	Nitrogen dioxide (NO_2)	1.0×10^{-7}
Helium (He)	0.00052	Nitrous oxide(NO)	6.0×10^{-8}
CH_4	0.00022	Sulphur dioxide (SO_2)	2.0×10^{-8}
Krypton (Kr)	0.0001	Hydrogen sulphide (H_2S)	2.0×10^{-8}
Di-nitrogen oxide (N_2O)	0.0001		

Structure of the Earth's Atmosphere

- The density and pressure of the atmosphere decrease with increasing altitudes.
- The atmosphere is divided into different layers based on temperature-
    - Troposphere
    - Stratosphere
    - Mesosphere
    - Thermosphere
    - Exosphere



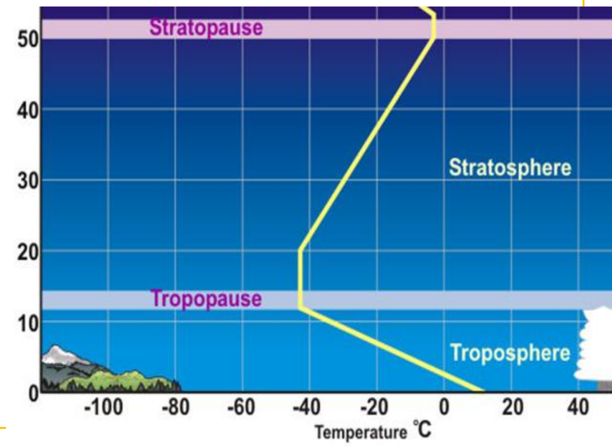
Structure of the Earth's Atmosphere

Troposphere – the lowest layer of the earth's atmosphere

- Extends from the earth's surface to about 12 or 15km
- Contains clouds, water vapour and dust
- Living organisms, air pollution, most aviation activities and all forms of weather occur here.
- Contains about 80% of the mass of the atmosphere
- The lowest part of the troposphere close to the earth surface is warmest. The temperature decreases with altitude.

Stratosphere– lies above the troposphere and separated by tropopause

- Extends from the top of the troposphere to about 50 or 55km
- Contains the ozone layer which absorbs UV radiations
- Thus, temperature increases with altitude
- It is almost free of clouds and weather patterns
- Most commercial airplanes (jet powered aircraft) fly in this region to avoid turbulences



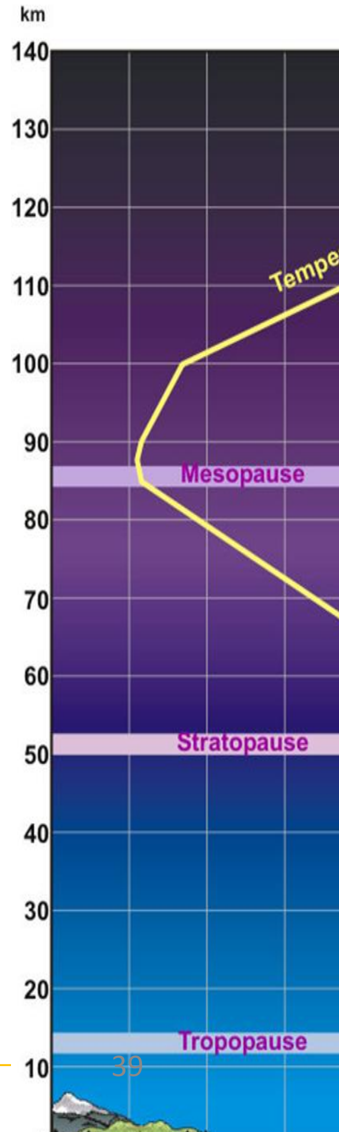
Structure of the Earth's Atmosphere

Mesosphere – the third layer of the earth's atmosphere extending from the stratopause to the mesopause to a height of about 80 to 85km

- The temperature decreases with altitude since there is no ozone and it's further from the earth's surface.
- The coldest temperatures (about -90°C) occur at the top of the mesosphere.
- Rocket-powered aircrafts fly in this region.

Thermosphere– lies above the mesopause to a height between 500 and 700km

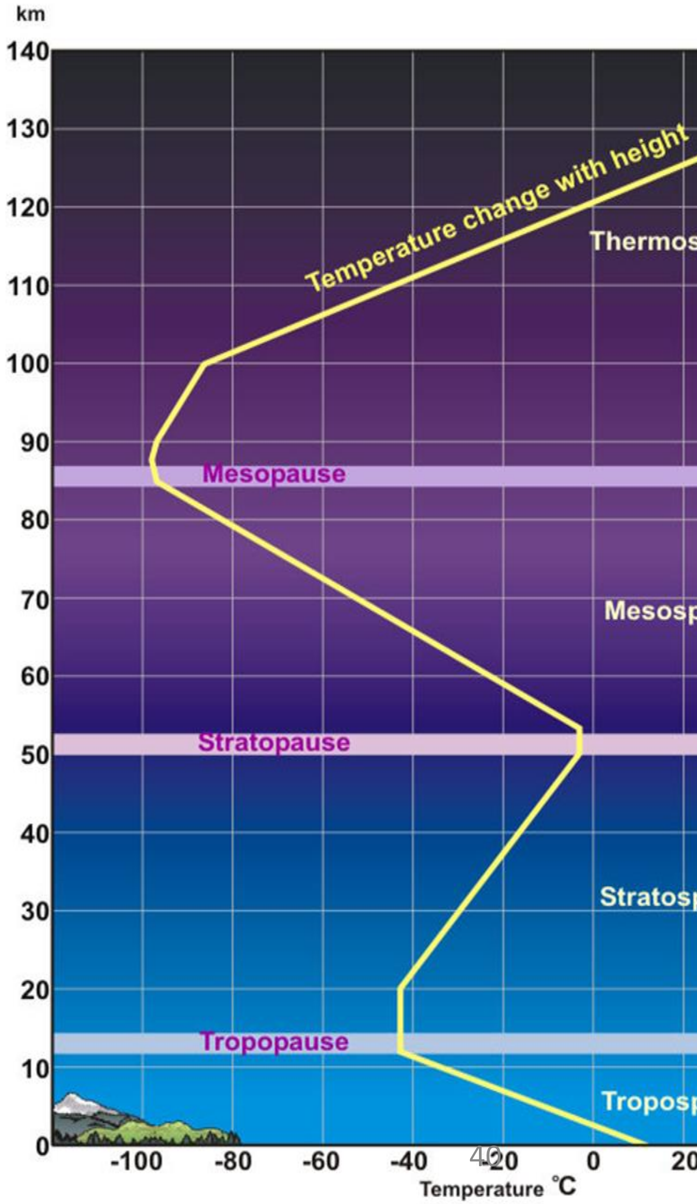
- Temperature increases with altitude due to absorption of gamma, UV and X-rays
- Very high temperatures- about 1500°C
- No clouds, water vapour and weather patterns
- Most satellites orbiting the earth occur here

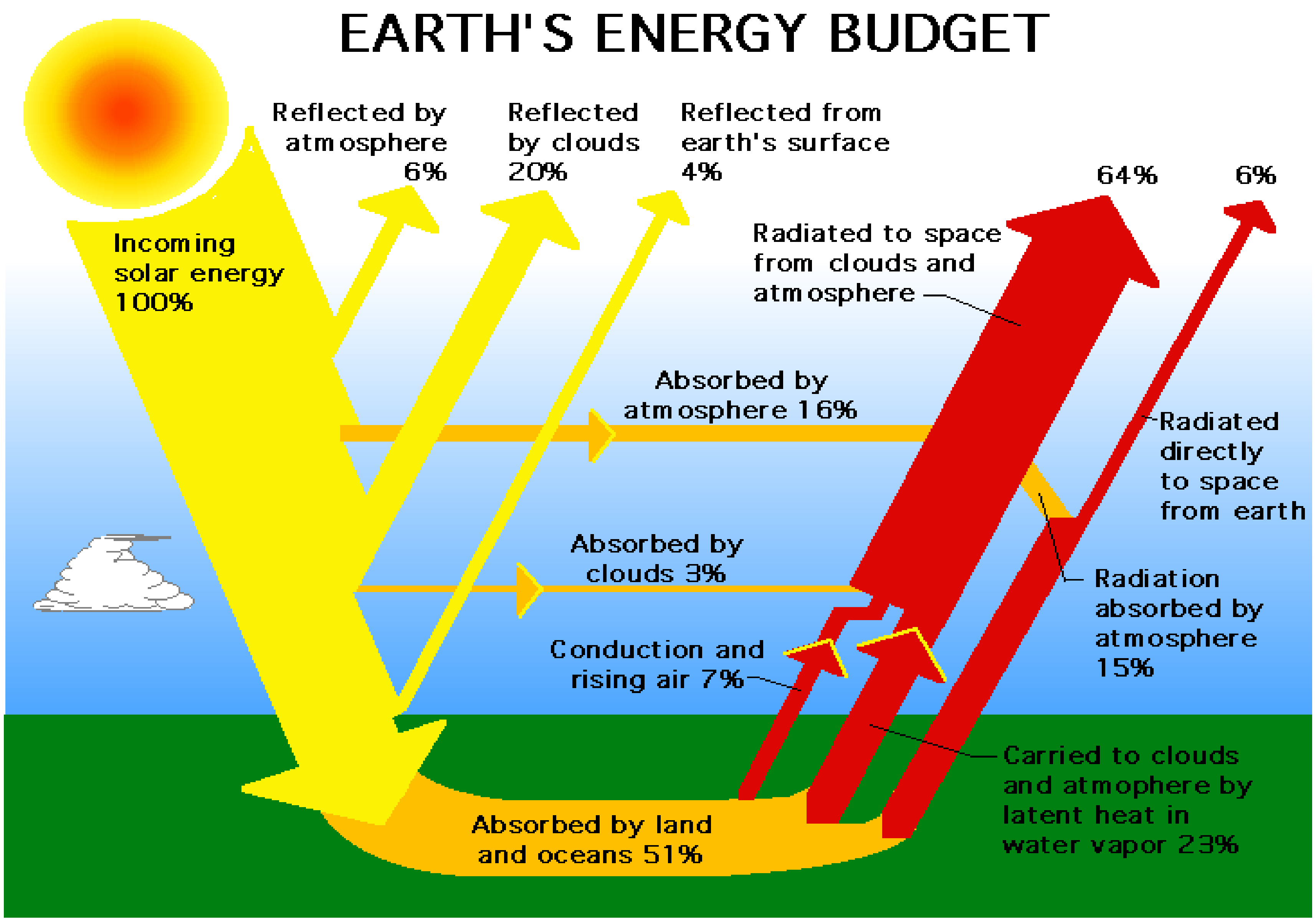


Structure of the Earth’s Atmosphere

Exosphere– the outer layer of the earth’s atmosphere extending from the thermopause to a height of about 700 to 10,000km

- The temperature decreases with altitude since there is no ozone and it’s further from the earth’s surface.
- Contains hydrogen, helium, nitrogen, carbon dioxide which are so widely apart, do not collide and often escape into space due to small force of gravity.
- Contains many artificial satellites moving round the earth.




EARTH'S ENERGY BUDGET
Reflected by atmosphere 6%
Reflected by clouds 20%
Reflected from earth's surface 4%
64%
6%
Incoming solar energy 100%
Radiated to space from clouds and atmosphere
Absorbed by atmosphere 16%
Radiated directly to space from earth
Absorbed by clouds 3%
Radiation absorbed by atmosphere 15%
Conduction and rising air 7%
Carried to clouds and atmophere by latent heat in water vapor 23%
Absorbed by land and oceans 51%

Emissivity and Albedo

Albedo: Is the reflection of solar radiation back into the atmosphere

Emissivity: The total amount of heat flow into space

FACTORS AFFECTING EMISSIVITY

- H_2O vapour [water vapour]
- CO [Carbon monoxide]
- CO_2 [Carbon dioxide]
- CH_4 [methane]

Human Impact on the Earth

How has man survived on the planet for the last hundred thousand years?

- Learning to live in the environment by co-operating with each other and through effective social organization
- Using language to increase the efficiency of co-operation and hand down knowledge of past survival experiences
- Belief and reliance on a Supreme Being in difficult and good times
- Learning to use tools for hunting, gathering and making food and protective clothing

Types of Human Societies and Their Impact

- There are different types of human societies based on the level of technology applied.
- These include:
  a. Early hunter-gatherer
  b. Advanced hunter-gatherer
  c. Pastoral society
  d. Horticultural society
  e. Agricultural society
  f. Industrial society
  g. Computer and robotics age
  h. Nanotechnology age

Types of Human Societies

Early hunter-gatherer

- This human group survived mostly by hunting animals, fishing and gathering plants
- Survival was based on their ecological knowledge- where and how to find which plant or animal.
- Nomadic- moving all the time in search of food and water
- Nature controlled them- e.g. availability of food determined their population size
- Material possession was minimal
- Used simple, crude tools and so could kill only few animals
- High level of interdependence -cooperation and sharing of resources
- Lived in small groups of not more than 50
- Belief in a Supreme Being as the Creator

Types of Human Societies

Advanced hunter-gatherer

- This human group survived mostly by hunting animals, fishing and gathering plants
- Survival was based on their ecological knowledge- where and how to find which plant or animal.
- Nature controlled them- e.g. availability of food determined their population size
- Lived affluent lives- more relaxed, less stress and anxiety
- Used more sophisticated tools e.g. stone axes, projectiles as well as fire to kill larger number of animals
- Their food was higher in quantity and quality
- High level of interdependence -cooperation and sharing of resources
- Effective social organization
- Small population groups which were widely scattered
- Had low adverse impact on the environment

Types of Human Societies

Pastoral society

- Humans started domesticating wild animals for food and transportation.
- Remained nomadic – search for fresh pasture
- Could produce surplus food for storage
- People could engage in non-survival activities such as trading and craftwork
- Had a more adverse impact on the environment than hunting society
- Environmental impact: increased water runoff, lowered water tables, erosion

Types of Human Societies

Horticultural society

- Developed almost the same time as pastoral society
- Humans started growing crops close to their homes using digging sticks and hoes.
- Thus, nomadism and varied diet were replaced by permanent settlement and monotonous meals
- Engaged in slash-and-burn agriculture- they burnt and cleared forests, used the ash to enrich the soil before growing crops
- After 2 to 5 years, farming lands are allowed to fallow. Farmers then move to cultivate new lands.
- Could produce surplus food for storage
- People could engage in non-survival activities such as trading and craftwork

Types of Human Societies

Agricultural society

- Humans used higher technology in growing crops and raising farm animals.
- Learned to use fire to clear land and practiced monocropping
- Thus, nomadism were replaced by permanent settlement
- Towns and cities developed
- They became shepherds of large flocks
- Landscape was changed dramatically
- Plant and animal diseases (Pests)
- Communicable diseases in man
- Water pollution
- Fought for land and water

Types of Human Societies

Industrial society

- Machines were made to manufacture goods and in agriculture
- Synthetic products to mimic natural things
- Water and Air pollution (acid rain)

Computer and Robotics Age :

- Fast transfer of information
- Rapid production, improvement and efficiency of delivery in the manufacturing industry
- Health problems which includes blood pressure, diabetes etc
- Increased organised crime

Nano age

Medical breakthroughs , communication, other impacts not yet known

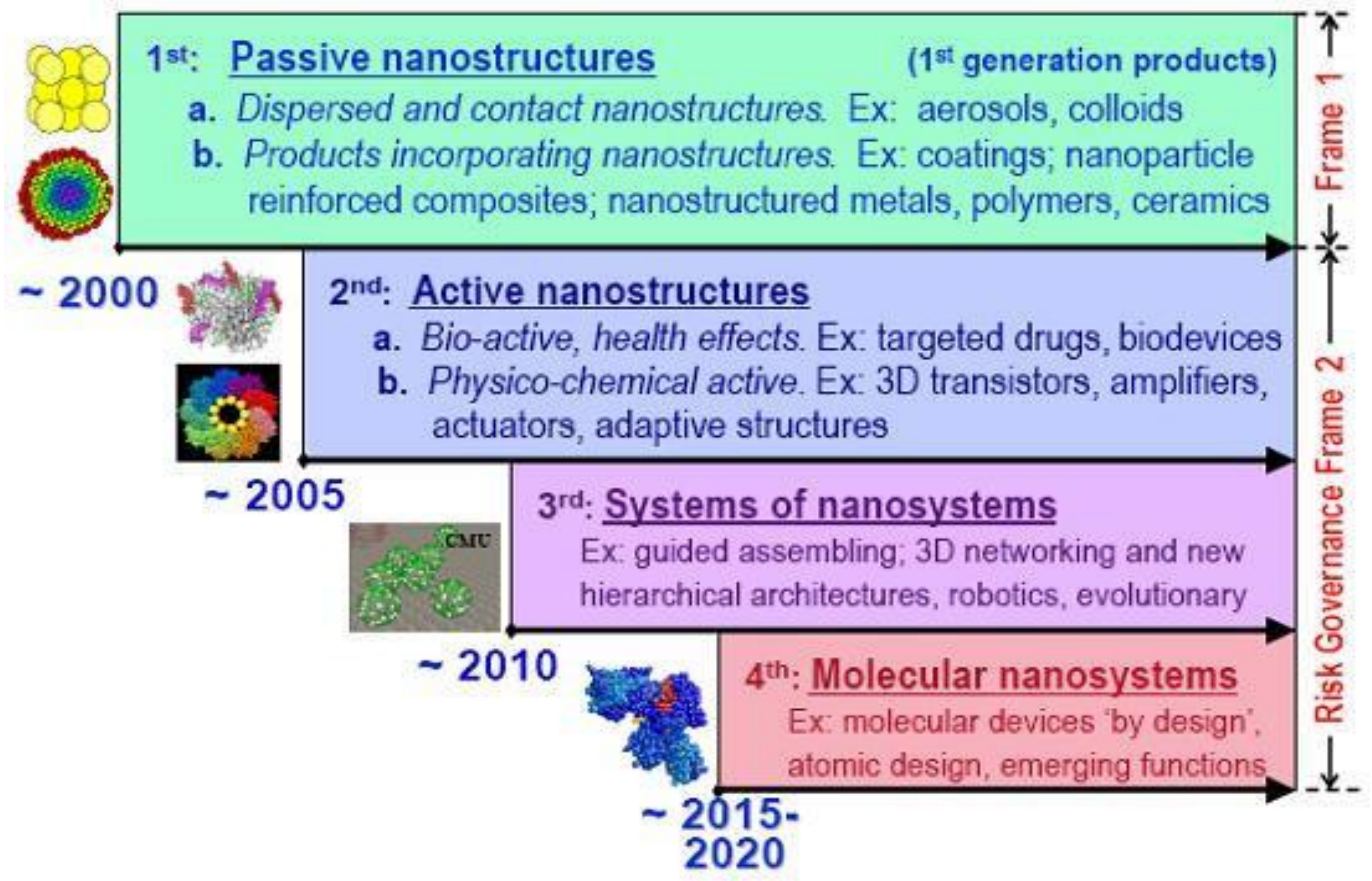
Nanotechnology

- This is the engineering of functional systems at the molecular and atomic level.
- It makes most products
  - lighter
  - stronger
  - cleaner
  - cheaper and less expensive.
  - Utilise less quantity of resources

More organised crime, Genetically modified foods, more air pollution problems

Holograms using light interference pattern to produce three dimensional images. This new technology is going to produce another society of its own

Nano structures



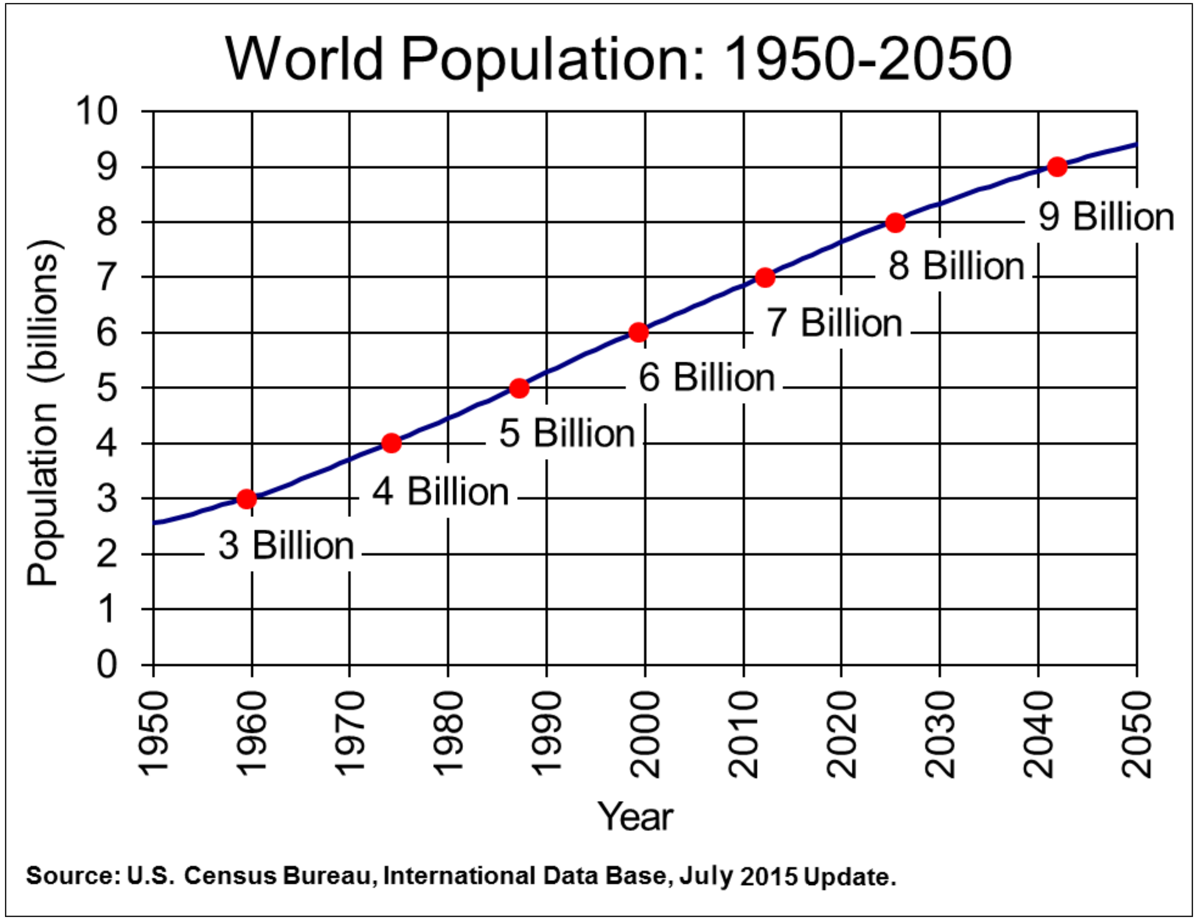
Human activities contribute heat to our environment

- Use of air condition in a house
- Use of washing machine
- Use of microwave oven
- Travels by any means using fossil fuel
- Cooking
- As more people live on earth they must be fed and clothed and all the activities to keep them alive produces heat which is increasing the temperature in the earth's atmosphere.
- Because with increase in population means increase in resource use and this is also becoming an issue to deal with
- Man has really had significant impact on the environment and he is having a back lash

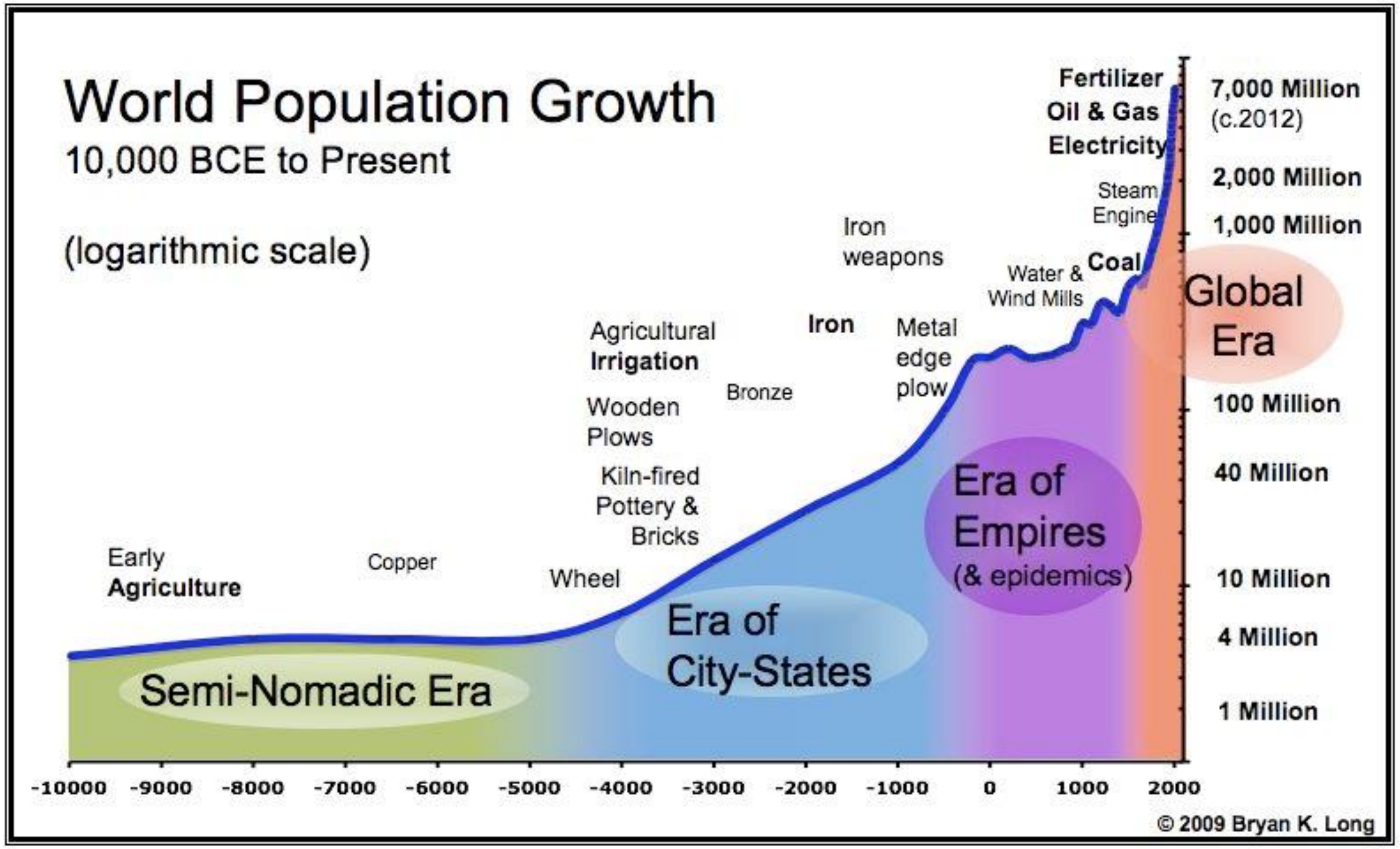
Population growth crises

- Population growth is the increase in the number of individuals in a population.
- Global human population growth amounts to around 83 million annually, or 1.1% per year.
- The global population has grown from 1 billion in 1800 to 7.774 billion in 2020.
- It is expected to keep growing, and estimates have put the total population at 8.6 billion by mid-2030, 9.8 billion by mid-2050 and 11.2 billion by 2100.
- Many nations with rapid population growth have low standards of living, whereas many nations with high standard of living have low population growth

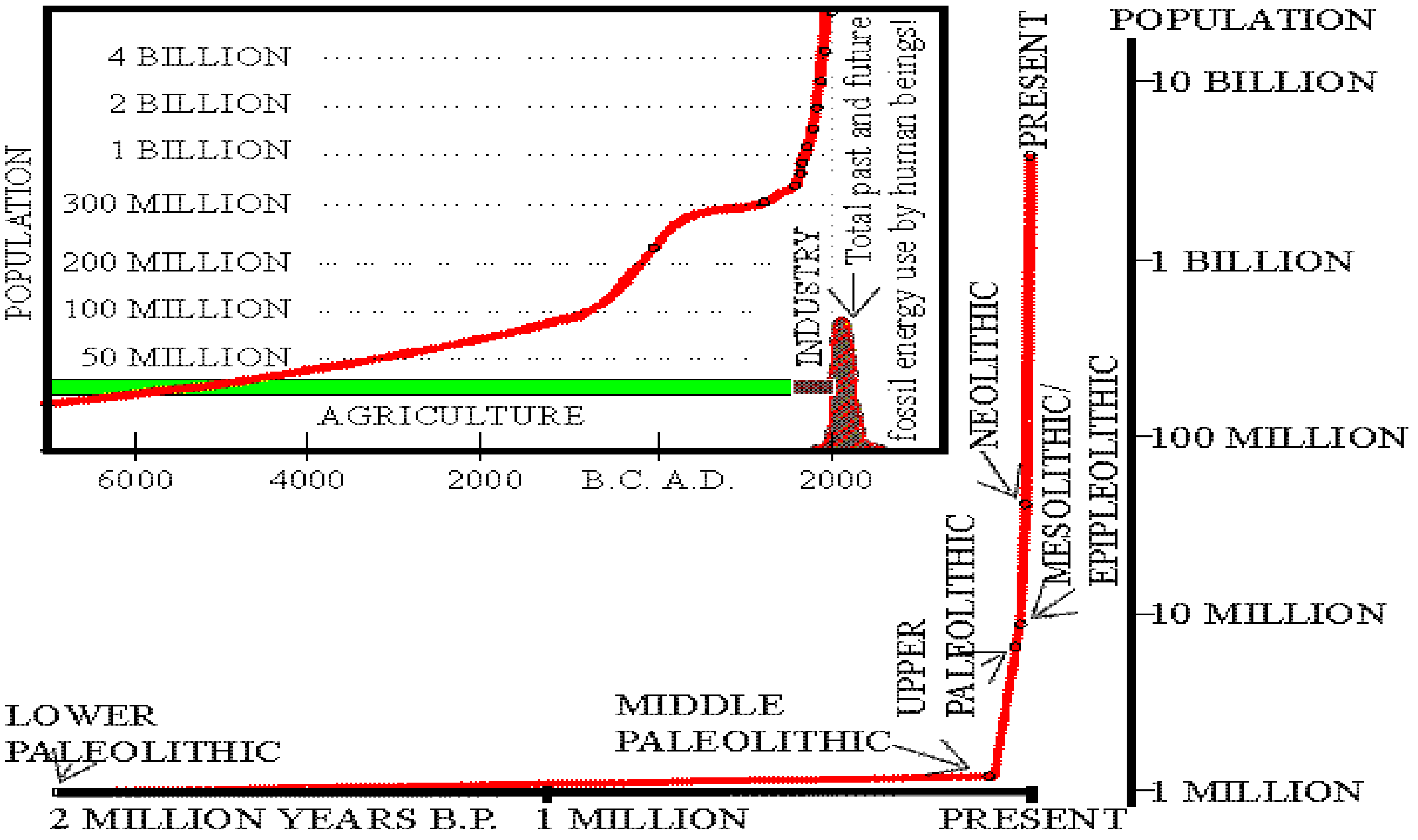
World population predictions



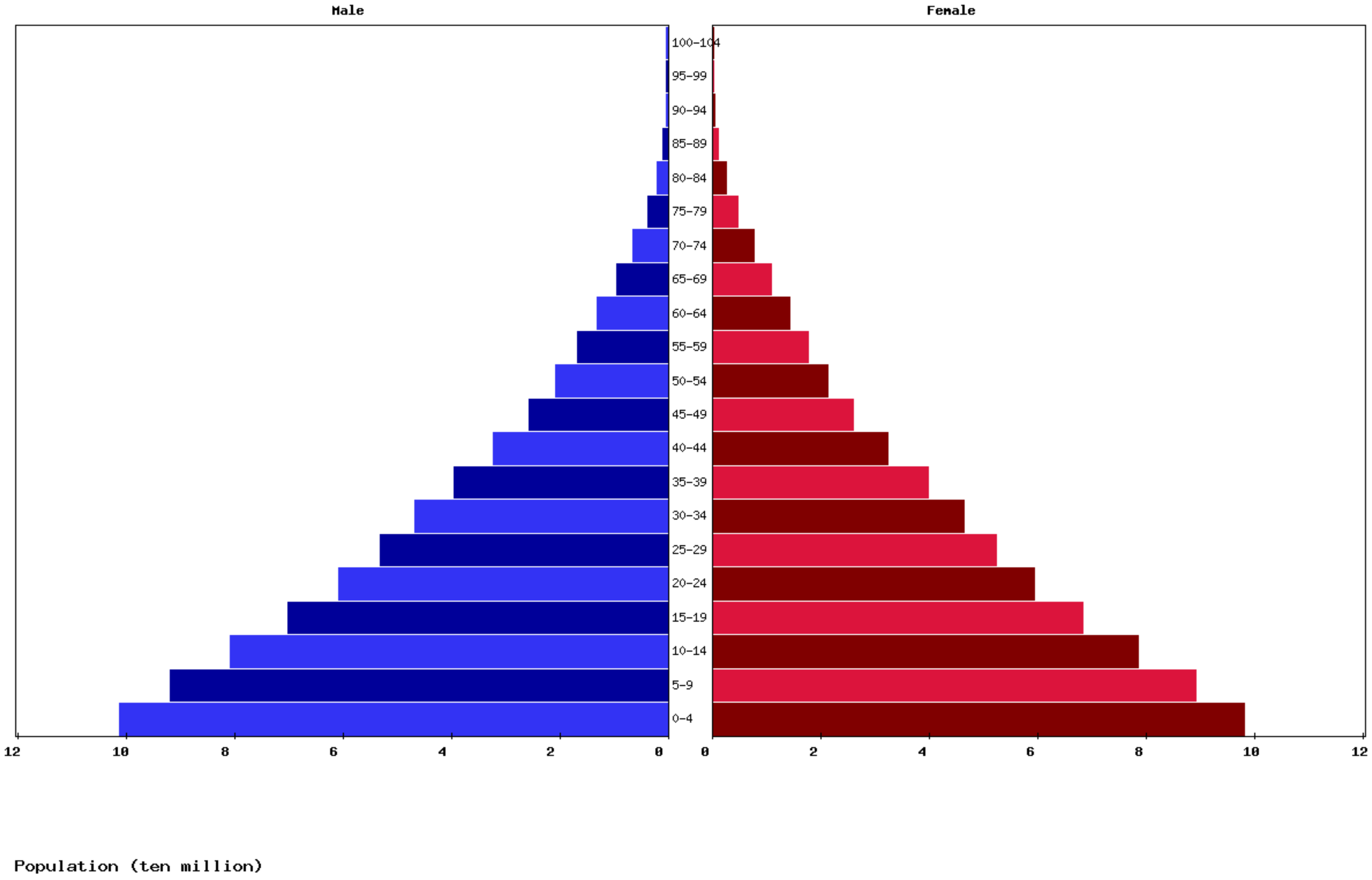
World population growth



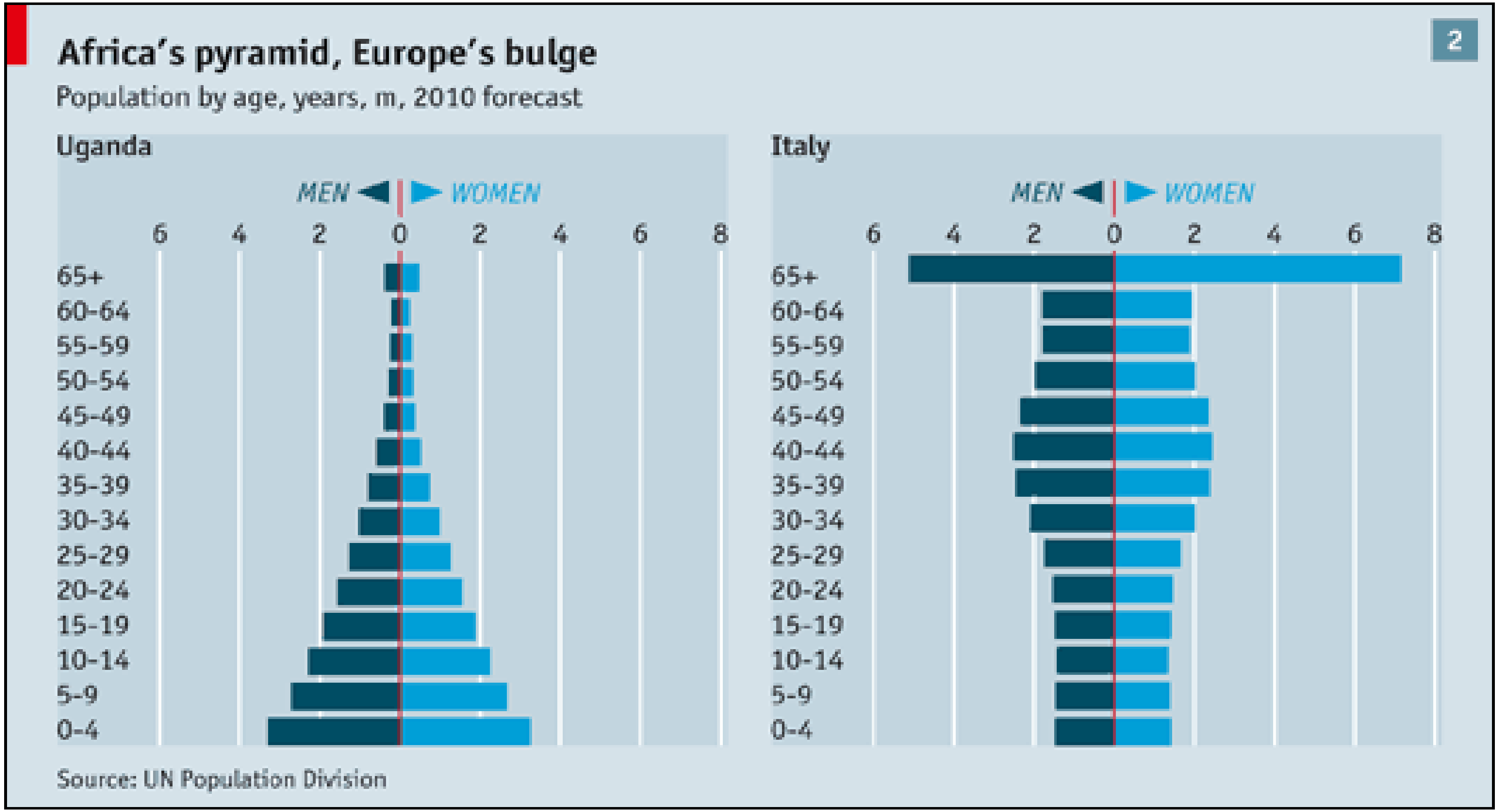
THE LOOMING CRISIS (J- CURVE)



Africa population pyramid



Comparison of population pyramids



Population versus resources

1. FACTORS INFLUENCING POPULATION GROWTH:

- Birth
- Immigration
- Food availabilty

2. FACTORS LIMITING POPULATION GROWTH

- Emigration
- Education
- Diseases

❖ If resources that support a population are depleted, this population could become extinct.

Definitions in population studies

- Growth rate -No of population increase within a year and is measured as number of birth minus the number of death.
- Crude birth rate – child births per 1000 population in a year
- Crude death rate- Number of death per 1000 people in a year
- Fecundity – Number of offspring one female can have under ideal conditions or as it exists in a particular community
- Real population growth rate takes care of immigration and migration within the year

Calculations for population growth

Population Growth Math

- Change in population = Births – Deaths
- Per capita birth rate = b
- Per capita death rate = d
- # of individuals = N
- Rate of population growth (r) = b – d
- Survivorship = % surviving

Ex: If there are 50 deer in a population, 13 die and 27 are born the next month. What is the population size the following month?
- *(Answer: 27-13 = 14, so new population is 64)*

Ex: What is the birth rate for the deer? *#Births/N = b*
- *Answer: 27/50 = .54*
- *Death rate (d) = 13/50 = .26*

Ex: What is the rate of growth for the deer? *r = .54 -.26 = .28*

AP Biology

The Earth's carrying capacity

- Carrying capacity is the maximum number of a species an environment can support indefinitely. Every species has a carrying capacity, even humans. However, it is very difficult for ecologists to calculate human carrying capacity. Humans are a complex species.
- Hunter gatherers' life style could support only 100 million
- Some experts are saying it is 40-50billion
- What will happen when we reach the earth's carrying capacity?
  - Increase in crime
  - Hunger
  - Pandemics
  - Wars
  - Natural disasters
  - Strife

Assignment 1

- What is the most important thing to prepare for as population increases?
- Has KNUST reached its carrying capacity?
- Suggest ideas and plans to avert the situation?
- What were the major impacts the agricultural society had on the environment?

Resource use and distribution has major impacts on the environments

- Understanding resources within our environment
- What is a resource?
- How human activities affect the quality, composition, constitution of these resources.

Resource

- Any physical or virtual entity of limited availability or anything used to help one earn a living.
- Anything needed by an organism or living things, a population or the ecosystem

Two divisions

- Natural Resources
- Human Resources

Natural Resources

- These are basically derived from the natural environment.
- Natural resources may be classified based upon

- Origin
- Stage of development
- Renewability

Origin

Biotic and Abiotic

Biotic – living things and materials derived from them e.g. Forest, animals, birds and their products

Abiotic – Non living resources e.g. land, water, air, minerals etc.

Stage of development

i. Potential resources

ii. Actual resources

Potential resources are;

- Those that exist in a particular region
- Their quantity are not known
- May be exploited and may be used in the future
- Eg. minerals may exist but if it has not be drilled and put to use, it remains potential
- High speed winds(some 200yrs ago)

Actual resources

- Resources that have been surveyed.
- Their quantity and quality have been determined
- They have being put to use in the present times
- Eg. Water, coal, petroleum

Renewability

- Renewable
- Non-renewable resources

Renewable

They can be used repeatedly and be replaced naturally. Can theoretically last for ever

e.g. the sun, plants and animals, fresh air, fertile soil, fresh water, energy from the sun and wind tides.

Non renewable sources

- Can be depleted after a period of time of exploitation.
- Recovery may be expensive and may be impossible e.g. metallic minerals e.g. gold, iron, copper, tin etc.
- Non-metallic e.g. fossil fuel, clay, sand, salt, phosphates etc.

Renewable and Non-renewable

- Renewable resources can become non -renewable resources if they are used at a faster rate than they can be replenished.
- Some animal and plant spp. have become extinct because the rate at which they are being utilised is faster than being replaced naturally
- The world is running out of resources e.g..

Fossil fuels - 50 yrs

Pb, Sn, Cu, Ag, Hg – 2000 – 2040 etc

Mitigation Measures for Resource Use Control

The 4Rs

- Recovery
- Recycle
- Reuse
- Reduction in consumption

- More devised technologies to recover more minerals from used ores;
- Alternative method or alternative materials for same purpose that a mineral or a particular resource is used.

Some Terminologies Relating To Environment Studies

- Environmental Control Systems (Engineering):

the system provides the occupants with a suitably controlled atmosphere to permit them to live and work in the area.

- Environmental ethics deals with:

the attitude of people towards other living things and towards the natural environment and engineers must have positive attitude toward the protection of the environment in their designs and operation of their systems

- Environmental Engineering
- Environmental Impact Assessment (EIA)

The Environment

1. The totality of influences acting upon an organism from without
2. The sum total of external conditions and influences affecting the development and life of an organism.
3. It is the aggregate of all natural and operational or other conditions that affects operation of equipment or components
4. Sum total of all external influence acting on the organism or on part of the organism
5. Is the aggregation of all conditions and the influence that determines the behaviour of a physical system

The Environment

- Physicochemical factors
  - That is factors that are physical and chemical conditions
  - Abiotic factors that affect the **environment**
  - Examples of **physicochemical** factors are: Temperature, relative humidity, light intensity etc

- Cultural: history (beliefs), attitudes and practices

- Socio-economic: population, economy, infrastructure

Environmental Ethics

- The Ethics is a branch of philosophy that seeks to define

what is right and what is wrong.

- The laws of any nation should match the ethical commitment of those living there.
- Not every action that is ethically right can have a law supporting it.
- E.g. 1) Not urinating/defecating in unauthorised place
- 2) Throwing rubbish in drains
- 3) Littering the street

Goal of environmental ethics

- The goal of environmental ethics is not just
  - To convince people to be concerned about the environment but;
  - To focus on the moral foundation of environmental responsibility.

Environmental Ethics Theories

- Three primary theories on moral responsibilities in relation to the environment.
- They include:

1) Anthropocentrism or human centred ethics
2) Biocentrism or life centred environmental ethics
3) Ecocentrism

Anthropocentrism

Anthropocentrism has the view that all environmental responsibility is derived from human interest alone.

It assumes that only humans are morally significant and have direct moral standing.

- In this view the adage is ‘protect when it benefits humans’. This view is flawed in the sense that ;
- The Earth remains environmentally hospitable for supporting human life
- It remains a pleasant place for humans to live now and for the future.

Biocentrism

- Biocentrism or life centred environmental ethics
- According to this theory all forms of life

Humans

animals

Plants

microorganisms etc.

- Have an inherent right to life.

Ecocentrism

- This approach / theory maintains that the entire environment deserves direct moral consideration.
- Not consideration that is derived merely from human or animal interests.
- In this regard everything existing on the Earth should have the same right to life as any other.

Environmental Ethics and Sustainable Development

Generally our attitudes or approaches to the environment must revolve around

- Development
- Preservation or
- Conservation.

For this to occur; our development should be sustainable.

Sustainable development (SD)

- SD is often defined as 'meeting the needs of current generations without compromising the ability of future generations to meet theirs'.
- SD focuses on the promotion of appropriate development to reduce poverty while still preserving the ecological health of the landscape.

Environmental Ethics and Sustainable Development

- SD hinges on three pillars namely: Economic development, Social development and Environmental protection.

❖At times consideration is given to cultural development as well.

Environmental Engineering

- Basically the application of theories of science and material forces to confront ecological and socio-economic problems so as to reduce pollution, contamination and deterioration of the surroundings in which humans live.

- Environmental Engineering involves control of water, soil and atmospheric pollution and the social and environmental impact of planned projects.

What Environmental Engineering Is

- The provision of solution of problems of environmental sanitation,

including

❖water and wastewater treatment to maintain good quality water,

❖ Clean contaminated Soils,

❖Reduce Atmospheric pollution, and

❖ Assess the social and environmental impact of these solutions and developmental projects such as bridges, road construction, provision of electricity for communities

Environmental Engineering ensures the following:

- Provision of safe water
- The proper disposal or recycling of urban rural waste
- The control of water, soil and atmospheric pollution
- The elimination of industrial and health hazards
- Provision of adequate sanitation in urban, rural and recreational areas

Assignment 2

- We are running out of resources. What practical ways would you recommend for packaging of milk and other canned foods?
- What are the Four Rs?
- What are pillars of sustainable development?
- Name the three theories of environmental ethics?
- Calculate the population of Kumasi for 2022 if the growth rate is 3% per annum and death rate is 1%. The impact of the opening of the borders to other Ecowas countries brings in 4% of immigrants from neighbouring countries. Residents leaving the Kumasi for greener pastures is also 0.05%
- What do environmental engineers do? (slides of this have been highlighted with more than two slides take note)